

test_type_matrix

Test-Type Matrix

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Audit of boba's actual test-type coverage against the §11.4.27 mandated list (unit / integration / e2e / full-automation / security / DDoS / scaling / chaos / stress / performance / benchmarking / UI / UX / Challenges / HelixQA). Produced as part of **BOB-074** (DDoS-class testing scaffold).

This is a **point-in-time audit** (2026-08-18) of directory/file presence and counts, not a coverage-percentage measurement (§11.4.224's code-coverage floor is a separate, unaddressed axis — see “Gaps” below).

Method

- challenges/scripts/*.sh — enumerated by filename + first-comment-line description.
- tests/ — enumerated by subdirectory (already type-organized) + file counts.
- download-proxy/tests/ — does **not exist**; per the project's own documented gotcha (“Merge service tests live at ./tests/, not download-proxy/tests/”), Python tests for the merge service live under the top-level tests/ tree instead.
- qBitTorrent-go/*/test*.go — enumerated by package + tests/{integration, e2e, security, contract} top-level suites.
- challenges/helixqa-banks/*.yaml — enumerated by bank + test-case count.

Coverage summary

§11.4.27 mandated type	Status (pre- BOB-074)	Status (post- BOB-074)	Where
unit	Present	Present	tests/unit/ (133 files), qBitTorrent-go/ internal/**/*_test.go (40 files)
integration	Present	Present	tests/integration/ (23 files), qBitTorrent- go/tests/integration/ (2 files)
e2e	Present	Present	tests/e2e/ (7 files), qBitTorrent-go/tests/ e2e/ (1 file), frontend/e2e/ (3 Playwright specs)
full- automation	Present (cross- cutting)	Present	No dedicated directory — governed as a discipline across every suite (§11.4.98); not independently re-audited here
security	Present	Present	tests/security/ (12 files), qBitTorrent-go/ tests/security/ (1 file), plus credential- focused challenges (credential_leak_grep_challenge.sh, cred_roundtrip_challenge.sh, master_key_autogen_challenge.sh, ...)
DDoS	ABSENT	Scaffolded	challenges/scripts/ ddos_resilience_challenge.sh (new, this commit) — see caveats below
scaling	ABSENT	ABSENT	No scaling-tagged directory, test file, or HelixQA bank anywhere in the tree. Followup filed.
chaos	Present (thin)	Present	tests/chaos/ (2 files), tests/stress_chaos/ (2 files), qBitTorrent-go/internal/api/ magnet_stress_chaos_test.go, qBitTorrent- go/tests/integration/ scheduler_hooks_sse_stress_chaos_test.go
stress	Present	Present	tests/stress/ (10 files), tests/load/ locustfile.py (Locust load profile)
performance	Present (thin)	Present	tests/performance/ test_concurrent_search.py (1 file), tests/ benchmark/ (3 files)
benchmarking	Present	Present	tests/benchmark/ (3 files + baselines/), Go *_bench_test.go (crypto_bench_test.go, autoconfig_bench_test.go)

§11.4.27 mandated type	Status (pre- BOB-074)	Status (post- BOB-074)	Where
UI	Present	Present	frontend/src/**/*.*.spec.ts (30 Vitest specs), frontend/e2e/*.*.spec.ts (3 Playwright specs)
UX	ABSENT (distinct)	ABSENT (distinct)	No test is framed around usability/accessibility/UX outcomes specifically; UI functional coverage exists (above) but nothing tags a UX dimension. Followup filed.
Challenges	Present	Present	challenges/scripts/*.sh — 26 scripts pre-BOB-074, 27 post (see enumeration below)
HelixQA	Present	Present	challenges/helixqa-banks/*.yaml — 7 banks, 61 test cases total; none tag DDoS or scaling (followup)

Gap count identified: 3 mandated types with zero representation at audit time — DDoS, scaling, UX. This task (BOB-074) scaffolds DDoS (partial close — see caveats). **scaling and UX remain fully open.**

Enumeration — challenges/scripts/*.sh (27 scripts)

Script	Primary type
boba_db_file_perms_challenge.sh	Security (file-permission gate)
credential_leak_grep_challenge.sh	Security (leak gate)
credentials_wired_challenge.sh	Integration / anti-bluff wiring guard
cred_roundtrip_challenge.sh	Integration (round-trip gate)
ddos_resilience_challenge.sh	DDoS (new, BOB-074)
docs_chain_verify_challenge.sh	

Script	Primary type
	Integration (docs/DB sync gate)
download_proxy_deep_challenge.sh	Unit/integration coverage gate
durable_run_helper_challenge.sh	Chaos/resilience regression guard
env_db_drift_challenge.sh	Integration (config drift gate)
envfile_bindmount_atomic_challenge.sh	Integration (atomicity)
helixqa_jackett_fake_behavioral_equivalence_challenge.sh	HelixQA equivalence guard
host_no_auto_poweroff_challenge.sh	Host-safety (CONST-033/034)
host_no_auto_suspend_challenge.sh	Host-safety (CONST-033)
install_sh_idempotent_challenge.sh	Integration (idempotency)
iptorrents_cookie_flow_challenge.sh	Integration/ security (cred flow)
jackett_autoconfig_clean_slate.sh	Regression guard (CONST-032)
jackett_cookie_login_hardening_challenge.sh	Security hardening guard
master_key_autogen_challenge.sh	Security (key-gen gate)
nmclub_native_plugin_clarification_challenge.sh	Integration clarification guard
no_suspend_calls_challenge.sh	Host-safety (source-tree gate)

Script	Primary type
private_tracker_html_challenge.sh	Unit (HTML-parsing coverage)
run_all_challenges.sh	Aggregator (not itself a test)
search_deep_coverage_challenge.sh	Unit coverage gate
status_docs_freshness_challenge.sh	Integration (doc-staleness gate)
tmux_survives_oond_pressure_challenge.sh	Chaos (§11.4.238 coverage-escape guard)
tracker_auth_live_challenge.sh	Integration (live auth)
upstream_proxy_wired_challenge.sh	Integration (cross-layer wiring)
workable_items_integrity_challenge.sh	Integration (SSoT integrity)

Enumeration — tests/ (Python, merge-search + shared)

Subdirectory	File count	Type
unit/	133	unit
integration/	23	integration
e2e/	7	e2e
security/	12	security
stress/	10	stress
stress_chaos/	2	stress + chaos (combined)
chaos/	2	chaos
concurrency/	3	concurrency (unmandated)

Subdirectory	File count	Type
		but useful — see note)
performance/	1	performance
benchmark/	3 (+ baselines/)	benchmarking
load/	1 (locustfile.py)	stress/load
property/	2	property-based (unmandated but useful)
contract/	4	contract (unmandated but useful)
observability/	2	observability (unmandated but useful)
docs/	3	doc-consistency (unmandated but useful)
hooks/	1 (test_guard_forbidden_commands.sh)	integration (git-hook guard)
fixtures/, memory/, test_torrents/	—	shared fixtures, not test files themselves

Note: concurrency/, property/, contract/, observability/, and docs/ are not literal §11.4.27 vocabulary items but are real, useful test disciplines this project has already adopted beyond the mandated floor.

Enumeration — qBitTorrent-go (Go)

Location	File count	Type
internal/**/*_test.go	40	unit
tests/integration/*_test.go	2	

Location	File count	Type
		integration (one is scheduler_hooks_sse_stress_chaos_test.go — chaos-tagged)
tests/e2e/*_test.go	1	e2e
tests/security/*_test.go	1	security
tests/contract/*_test.go	1	contract
internal/db/ crypto_bench_test.go, internal/jackett/ autoconfig_bench_test.go	2	benchmarking
internal/api/ magnet_stress_chaos_test.go	1	stress + chaos (combined)

Enumeration — challenges/helixqa-banks/*.yaml

Bank	Test cases
boba-boba-ctl.yaml	7
boba-bobalink.yaml	12
boba-docs-chain.yaml	3
boba-download-proxy.yaml	10
boba-frontend.yaml	10
boba-services.yaml	14
boba-tmux-session-hardening.yaml	5
Total	61

None of the 7 banks tag a DDoS or scaling scenario.

Enumeration — frontend/ (Angular 21, UI)

Location	File count	Type
src/**/*.spec.ts	30	UI unit (Vitest)
e2e/*.spec.ts	3	UI e2e (Playwright)

Gaps + followups filed

Per §11.4.6 (no invented thresholds/fixes) these are filed as **followup work items**, not silently fixed inside this task's scope (challenges/scripts/ddos_resilience_challenge.sh + docs/testing/ddos_resilience.md + this file only):

1. **Scaling-class testing is fully absent.** No test, script, or HelixQA bank distinguishes “scaling” (does the system handle a growing dataset / tracker count / concurrent-user count gracefully, e.g. horizontal scale-out of the merge-search fanout, DB growth under challenges/helixqa-banks/boba-services.yaml's tracker set, or the qbittorrent-proxy-go --profile go swap) from “stress” (does it survive a burst). **Followup filed as workable item BOB-109** — “Scaling-class test coverage absent from boba's mandated test-type matrix” — scoped to at least one scaling dimension (e.g. tracker-count scale-out in merge search) with a real, measured baseline.
2. **UX-class testing has no distinct representation.** UI functional coverage exists (Vitest + Playwright) but nothing is framed around usability/accessibility/UX outcomes (e.g. WCAG checks, keyboard-nav coverage, screen-reader labeling). **Followup filed as workable item BOB-110** — “UX-class test coverage (accessibility/usability) absent” — scoped to an axe-core or equivalent accessibility pass over the Angular frontend.
3. **DDoS-class testing was fully absent; now scaffolded, not fully closed.** See docs/testing/ddos_resilience.md “Followups” for the two concrete items this scaffold surfaced:
 - No rate limiting exists anywhere in boba's stack (verified 2026-08-18: no slowapi/limiter/throttle in the Python service, no rate-limit middleware in either Go service, no nginx/reverse-proxy layer). **Followup filed as workable item BOB-111** — configure a real rate limit for the three public endpoints (candidate approaches documented in docs/testing/ddos_resilience.md).
 - boba-jackett's /healthz (qBitTorrent-go/internal/jackettapi/health.go:60-63) makes a synchronous, uncached Jackett.GetCatalog() call on every hit — under a cold-start concurrent burst this was measured to cause up to 98/150 (65%) of health-check requests to time out at 3s, recovering to <50ms/request once the burst subsided. This is a genuine self-inflicted amplification vector: an attacker (or even a mis-configured monitoring probe hitting /healthz too aggressively) could make the Jackett-management API's own health surface appear down

without ever touching Jackett itself. **Followup filed as Bug workable item BOB-112** citing this exact reproduction + recommended fixes (cache the Jackett liveness signal with a short TTL refreshed by a background ticker; add a tight timeout/circuit-breaker around the GetCatalog call so /healthz itself never blocks past ~250-500ms regardless of Jackett's state). See docs/testing/ddos_resilience.md "Findings" for the full evidence.

- The real fanout path (POST /api/v1/search, up to 43 real trackers) is deliberately **never** driven by the new challenge — see docs/testing/ddos_resilience.md "Scoping decisions" for why, and the followup this leaves open (a sandboxed/mock-tracker DDoS variant that exercises the orchestration code path without touching real third-party tracker sites).
4. **§11.4.224 code-coverage floor** ($\geq 85\%$, ~100% target) is a distinct, unaddressed axis from this test-type audit — this document counts *presence* of test types, not *coverage percentage* within them. Not remeasured here; out of this task's scope.

These followups were originally **documented here and in the DDoS doc rather than inserted into docs/workable_items.db directly** — this task ran in parallel with other subagents also touching shared project state, and a concurrent SQLite write from this task risked a race with theirs. The orchestrating session registered the followups above as tracked workable items (§11.4.93/§11.4.202) once the parallel batch completed: scaling → **BOB-109**, UX → **BOB-110**, rate limiting → **BOB-111**, /healthz amplification Bug → **BOB-112**, wrk dev-tooling gap → **BOB-113**, and the rate-limit detector's missing self-validation golden-bad fixture (see docs/testing/ddos_resilience.md "Findings" item 2) → **BOB-114**. The sandboxed/mock-tracker POST /api/v1/search fanout extension (this section's item 3, third bullet) remains an open scoping note, not filed as a separate BOB-NNN this round.